

# 1 Weeks 5-7: topics covered in lecture and assignments

- weeks 5-7
  - Desnanot-Jacobi identity for determinants; condensation.
  - Gessel-Viennot matrix, nonintersecting lattice paths.
  - Transfer matrix method for enumerating statistical-mechanical configurations.
  - Partially ordered sets; linear extensions, rank functions, topological sorting.
  - Random generation of reduced words (naively, recursively using the descent set)
  - Aztec Diamond and the domino shuffling algorithm
- weeks 8-10
  - Partially ordered sets, rank functions, linear extensions, chains.
  - Topological sorting
  - Loop Erased random walk, Wilson's algorithm
  - Plain Changes
  - Markov chains and random walk, irreducibility, steady state distribution
  - reversibility of markov chains, detailed balance condition
  - ising model, glauher dynamics

So, in addition to the problems below (and maybe more importantly), make sure you know all of the definitions of the above things, and have reviewed the assignment problems and in-class coding exercises that we did.

## 2 Problems from weeks 5-7

1. Verify that there are 175 pairs of nonintersecting paths which start at the points (0,1) and (1,0), end at the points (3,4) and (4,3), and take steps either one unit up or one unit right.
2. Find all rectangular standard Young tableaux with 2 rows and 3 columns. Then, find all topological sorts which correspond to these tableaux.
3. Define the descent set of a permutation. Explain why all reduced words for that permutation must end with a letter from the descent set.
4. Consider a Kasteleyn matrix of the order  $n$  Aztec diamond graph, all of whose edges have weight 1. Explain why the entry of  $K^{-1}$  corresponding to the edge at one of the "corners" of the diamond is  $\pm(1 - \frac{1}{2^{n-1}})$ . Hint: don't try to invert the matrix... instead, what is the probabilistic question I'm asking?

5. Let  $d_{-k}, d_{-k+1}, \dots, d_0, \dots, d_k$  be constants, and let  $M(d_{-k}, \dots, d_k)$  be the matrix whose  $(i, j)$  entry is  $d_{j-i}$  ( $M$  is called a Toeplitz matrix). What does the Desnanot-Jacobi identity say for  $\det M$ ?
6. Let  $P$  be the  $n$ -cube poset,  $(0, 1)^n$ , with product order. Compute the rank function of  $M$ .
7. Give an example of a poset that is not ranked.
8. Give an example of a ranked poset that is not a lattice.
9. Write down the row-to-row transfer matrix for the dimer model on a  $2 \times n$  grid, with constant weights. This is a matrix whose rows and columns represent all ways a row of vertices in a grid could be covered by dimers, and whose  $i, j$  entry is 0 or 1 according to whether configuration  $i$  can sit beside configuration  $j$  as two adjacent rows. How would you use  $M$  to compute the partition function of the dimer model on this grid?
10. Verify, either by hand or using the transfer matrix, that the number of domino tilings on a  $2 \times n$  grid is  $f_{n+1}$ , the  $(n + 1)$ th fibonacci number.

### 3 Problems from roughly weeks 8-10

1. Describe the row-to-row transfer matrix for the Ising model on a  $2 \times n$  grid, and explain how to use it to compute the partition function of the Ising model.
2. Give an example of a markov chain which is irreducible and aperiodic, but not reversible.
3. Let  $M$  be an irreducible markov chain. Explain why the periods of any two states of  $M$  are the same.
4. Consider the following Markov chain. The state space is the symmetric group  $S_n$ . At time  $t$ , we multiply by a randomly chosen simple transposition (i.e. the permutation that swaps  $i$  and  $i + 1$  and leaves everything else fixed, where  $i$  is a uniformly random number between 1 and  $n - 1$ ). The convergence theorem doesn't apply to this markov chain - why? Modify this markov chain so that its unique steady state distribution is the uniform distribution on permutations.
5. Suppose you have a large graph  $G = (V, E)$ , and  $u, v$  are two of  $G$ 's vertices. Let  $P$  be the path that joins  $u$  to  $v$  in a random spanning tree. What is the probability distribution of  $P$ ?
6. Write down plain changes for  $S_4$  (i.e. list the permutations in  $S_4$  in an order such that two adjacent permutations differ by a simple transposition).